Mohamed Abdelmonsef



Web Developer & Competitive Programmer

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Summary

High-achieving junior at Red Sea STEM High School (Class of 2028) with a strong focus on computer science and mechanical engineering. Passionate about applying academic research to real-world problems, demonstrated through engineering the ROTOR X wind turbine prototype and developing STEM Guide and Atlas platforms. Eager to contribute technical skills and a collaborative mindset to rigorous academic programs.

Work Experience

Web Development Experience

Lead Full-Stack Developer | 2026 - Present

- Developed **Atlas**, a web-based programming learning platform, utilizing Next.js and Tailwind CSS with dynamic routing and responsive layouts.
- Engineered **stemguide.org**, an educational platform for STEM students in Egypt, managing domain registration, Zoho workspace setup, and student outreach.
- Maintained comprehensive repository documentation and source control via GitHub to ensure clean, scalable code.

Hackathon & AI Experience

AI Developer & Global Finalist | May 2026 – June 2026

- Competed in the high school track of the **USAII global AI hackathon**, developing an AI-driven wildfire detection system to combat climate change.
- Collaborated with a team to conceptualize and pitch the machine learning model, reaching the global finals.

Engineering & Research

Engineering Researcher | 2025 – Present

- Designed the **ROTOR X** micro-scale horizontal-axis wind turbine, utilizing gear laws, NACA airfoils, and structural optimizations for clean energy generation.
- Conceptualized **Corrobot**, a planned AI-integrated robotic system designed to detect and resolve structural metal corrosion on bridges and turbines.

Competitive Programming

Competitive Programmer | 2024 – Present

- Trained rigorously for the Egyptian Collegiate Programming Contest qualification rounds.
- Implemented complex algorithms and data structures using C++ to solve timed, advanced problem sets on Codeforces.

Education

Red Sea STEM High School

High School Diploma | Expected Graduation: July 2028

- GPA: 3.85 / 4.00
- Capstone Project (ROTOR X): Engineered a micro-scale horizontal-axis wind turbine utilizing NACA airfoils and custom gear laws to optimize clean energy generation and aerodynamic efficiency.

Additional Information

- **Technical Skills:** Web Application Development, Software Architecture, and Database Design.
- **Soft Skills:** Analytical Thinking, Problem Solving, Cross-functional Leadership, Technical Communication, Stakeholder Management.
- **Languages:** English, Arabic, French
- **Programming Languages:** Python, C++, JavaScript, Wolfram
- **Web Technologies:** HTML5, CSS3, Responsive Design
- **Computer Science:** Data Structures, Algorithms, Object-Oriented Programming(OOP), Problem Solving.